**VBLS — $\alpha$ sensitivity: non-monotonic $t_B(\alpha)$ with optimum $\alpha^* \approx 0.10$ yr$^{-1}$**

**Active regime ($\alpha < \alpha^*$):** $\alpha \uparrow \Rightarrow \eta_{abs} \downarrow \Rightarrow F_{diss} \downarrow \Rightarrow \Sigma \downarrow \Rightarrow t_B \uparrow$ | **Collapse regime ($\alpha > \alpha^*$):** $\eta_{abs}$ **collapses,** $E \to E^*$ **early** $\Rightarrow t_B \downarrow$










